



The ties that transplant: The social capital determinants of the living kidney donor relationship distribution

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ABSTRACT

The network perspective on social capital decomposes it into ego's network size, alters' relevant resources, and social factors moderating access to alters' resources, but rarely examines how it is distributed across relationship types. Using this approach, I investigate the situationally-relevant social capital relationship distribution and its association with health-related social support, with an application to the living kidney donor relationship distribution. Analyzing an original survey of transplant candidates ($N = 72$) and their reports on their family and friends ($N = 1548$), I compare the tie count, donation-relevant biomedical resource, and tie strength relationship distributions to administrative data on the national distribution of living kidney donor relationships. I find that the tie strength relationship distribution matches the completed living kidney donor relationship distribution far better than the tie count and donation-relevant biomedical resource relationship distributions. These conclusions are upheld in race- and gender-stratified analyses and are robust across alternative approaches.

1. Introduction

Social capital is now well-established as an important determinant of individual and collective health, as research in this area has grown dramatically in recent decades (Chapman et al., 2022; Moore and Carpiano, 2020). Although social capital and health research adopts a variety of empirical and conceptual approaches (Fulkerson and Thompson, 2008; Portes, 1998), in its 'network perspective' incarnation, this literature brings focus to the health-promoting resources that inhere in individuals' connections to others, which may or may not be converted to actual social support in times of need (Portes, 1998; Carpiano, 2006; Lin, 1999, 2001; Moore et al., 2006; Song, 2011; Song and Lin, 2009). In this framework, ego's (the focal individual's) social capital endowment is the joint product of network size, situationally-relevant resources, and tie strength that moderates access to those resources.

The living kidney donor relationship distribution (i.e., the distribution of living kidney donation across relationship groups) is a particularly apt application of these theoretical concerns. Living donor kidney transplantation is the optimal therapy for end-stage kidney disease (i.e., kidney failure) (Lentine et al., 2023). Living kidney donors are typically members of the recipient's social network, with spouses/partners, siblings, children, friends, and parents collectively accounting for about 85% of living kidney donors in recent years (calculations described below). It is unclear, however, whether these relationship groups' high living kidney donor concentration is the result of their high social capital concentration and, if so, which social capital dimensions best explain this result. Corresponding to the network perspective on social capital theory, the living kidney donor relationship distribution could be generated through a combination of three social capital relationship distributions (i.e., the distribution of social capital characteristics across

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relationship groups): (1) the network tie relationship distribution (2) the transplant-related resource (i.e., donation-promoting biomedical characteristics), relationship distribution, and (3) the tie strength relationship distribution (Daw and Verdery, 2023). Additionally, due to large racial/ethnic and gender disparities in living donor kidney transplantation (Lentine et al., 2023), it is important to determine whether the findings differ appreciably by race and gender.

The answer to these questions is critical given the 15-year decline in living donor kidney transplants 2004–19, particularly among non-White transplant candidates (Rodrigue et al., 2013; Al Ammary et al., 2019, 2020), with the result that $\leq 20\%$ of non-Hispanic White transplant candidates and $\leq 10\%$ of non-White kidney transplant candidates receive a living donor kidney transplant within 2 years (Purnell et al., 2018). If underutilized relationship groups' donation rates are inhibited by their poor health or low numbers, these barriers to increasing their share of living kidney donors may be largely insurmountable in the medium term. However, if tie strength barriers appear to be the primary inhibitor (or if no social capital characteristic appears to be), this suggests that effective, ethical social and procedural interventions to promote living kidney donation in these groups are possible.

However, substantial data limitations inhibit these research objectives. Thanks to national transplant registry data (described below), we have records of every living donor kidney transplant conducted in the U.S. for the last few decades, including the donor-patient relationship. Accordingly, describing the living kidney donor relationship distribution is a straightforward enterprise. What is far less straightforward is understanding the social capital processes that drive the living kidney donor relationship distribution. The data limitation at work here is a more general one: in order to understand the determinants of a behavior, you need information not only on those who engage in that behavior, but also on those who were at risk of engaging in that behavior without doing so. Because members of transplant candidates' social networks are an overwhelming share of living kidney donors, we can treat this group as the population at risk of living kidney donation. In an ideal scenario, we would measure transplant candidates' full social networks regardless of living kidney donation and seek to explain donation as a function of network members' relationship, social capital, and individual characteristics. Unfortunately, no current dataset can directly link transplant candidates' social networks to completed living kidney donation.

In this paper I adopt a novel, indirect, and descriptive approach to this class of social network data limitations. The key idea is that although we lack data about kidney transplant candidates' social network members who do not become living kidney donors, we can gather these data on a sample of the kidney transplant candidate population. We can then compare the characteristics of each relationship group's *share* of social capital factors in the sample data (network ties, strong ties, well-resourced ties) to the same relationship group's *share* of living kidney donors in the registry data. This approach answers a subtly different question than that answered in the ideal data scenario: instead of asking what factors predict individual donation, here we will ask *why some relationship groups are more commonly represented among living kidney donors than other relationship groups*. In this way, we can compare how well the *social capital relationship distribution* corresponds to the *living kidney donor relationship distribution*.

Specifically, I employ scatterplot analyses to compare registry data on the U.S. living kidney donor relationship distribution 2008–17 to each social capital dimension's relationship distribution in a separate survey sample of 72 kidney transplant candidates and their reports on 1548 social network members. I find that the tie strength relationship distribution in survey data accords most strongly with the living kidney donor relationship distribution from registry data, while other social capital dimensions' relationship distributions are largely unrelated to the living kidney donor relationship distribution. This result is replicated in race- and gender-stratified analyses. These results also highlight key relationship groups such as cousins and nieces/nephews whose numbers and transplant-related resources could support higher than observed living kidney donation levels. Interventions to ethically promote higher living kidney donation levels in these groups should be explored.

2. Social capital, social networks, and living donor kidney transplantation

2.1. The network perspective on social capital and social support

Numerous theoretical and empirical approaches have been adopted in the study of social capital and health. My framework for social capital most closely follows that of Lin and colleagues (Lin, 1999, 2001; Song and Lin, 2009; see also Daw and Verdery, 2023). In this approach, social capital is defined as resources latent within individuals' social networks, which may or may not be mobilized into social support depending on situational need and ego-alter relational characteristics. In this approach, ego's endowment of social capital is a joint function of three key, interrelated traits. First, the amount of capital in ego's network or relationship group will, all else equal, be proportional to the *number of ties* in the network or relationship group. For instance, if all alters had equal resource and tie strength levels, more numerous relationship groups would be expected to provide more social support. Second, alters will vary in their *situationally-relevant resource* endowments. The situational-relevance of these resources is important because, while many resources such as money and cultural capital may be useful in a wide variety of contexts, other resources may only be relevant in particular situations. Third, relational characteristics linking ego and alter such as *tie strength* modify ego's access to alters' situationally-relevant resources. For instance, if all relationship groups were equally-numerous and well-resourced, the strongest tie relationship groups would be expected to provide more social support.

This network perspective differs significantly from other approaches to research on social capital. For one example, this perspective differs from communitarian approaches to social capital and health research, which are focused on social cohesion and civic engagement (Kawachi et al., 1997; Putnam, 2001), because the present focus is on properties of individuals' social networks, not communities. The network perspective also differs somewhat from social support-based definitions of social capital, which refer to the specific forms of assistance individuals render to one another, such as emotional, financial, and instrumental support. Instead, this paper's approach follows Lin, 1999, 2001 by treating them as related but distinct theoretical constructs. Instead, I investigate which

social capital dimensions' relationship distributions best explain social support relationship distributions. Thus, each social capital dimension is treated as a key predictor of social support relationship patterns.

Previous research on health-related social support provision has provided significant insights into its distribution across relationship groups, and how these patterns vary by the type of support being studied (House et al., 1988a, 1988b; Thoits, 2011). However, the health-related social support relationship distribution is rarely considered alongside its underlying social capital relationship distribution. This state of affairs is largely attributable to data availability; since most studies employ name generators that solicit names of people that provide different types of social support (Perry and Pescosolido, 2015), we often know much less about alters who do not provide social support, and therefore cannot test whether social capital and its dimensions differentiates relationship groups that do and do not provide high social support levels.

2.2. Social capital and the living kidney donor relationship distribution

This paper examines the correspondence of each living donor kidney transplant (hereinafter, LDKT)-related social capital dimension to the living kidney donor (LKD) relationship distribution. In this approach, LKD is construed as a highly-important form of social support nearly always provided by members of the kidney transplant candidate's family and social network (Lentine et al., 2023). The situationally-relevant social capital relationship distribution will consist of a joint function of the relationship distributions of three characteristics: the share of ties in each relationship group, the share of ties with transplant-related biomedical characteristics in each relationship group, and the share of strong ties in each relationship group.

Fig. 1 provides an overview of the hypothesized relationship distribution of each social capital dimension as related to LKD, where darker shading in each panel indicates a higher share of social capital. The following sections explain the relevance of each characteristic and the evidence behind their hypothesized distributions, relating each literature to the corresponding Fig. 1 panel.

2.2.1. The social network tie relationship distribution

The raw social network tie relationship distribution is the simplest potential explanation for the LKD relationship distribution. Although relationship groups' share of network ties will vary considerably between individuals and across age groups, on average, extended kin such as cousins, aunts/uncles, nieces/nephews, and grandchildren are more numerous than direct, living ancestors (parents, grandparents), direct descendants (children), or siblings and spouses/partners (Daw et al., 2016; Margolis and Verdery, 2019; Murphy, 2011). Additionally, friends are likely among the more numerous network ties. Research finds that friends were the most common type of important matters confidants (McPherson et al., 2006), though others have critiqued this work as methodologically flawed (Fischer, 2011; Paik and Sanchagrin, 2013). Similarly, though, national polls find that about 40% of U.S. adults in 2003 had 6 or more good friends, while fewer than 20% had 0-2 good friends (Fischer, 2011). Additionally, there is some evidence that kinship structures differ moderately by race due to differential rates of mortality, marriage, and fertility (Daw et al., 2016; Verdery and Margolis, 2017; Martin, 2020; Mayol-García et al., 2021; Patterson et al., 2020). In contrast, gender groups are unlikely to differ considerably in the structure of their kinship networks since the sex (and by correlation, gender) of each individual birth is largely unrelated to the distribution of kinship ties, and there is little evidence of gender differences in numbers of friends (McPherson et al., 2006).

Panel A in Fig. 1 depicts the hypothesized tie count relationship distribution. In this panel, relationships labeled 'restricted' are those where the potential tie count is biologically limited within the dominant American kinship system (i.e., one spouse/partner, two biological parents, and four biological grandparents, potentially with additional ties through ex-, step-, and adoptive means). Those labeled 'less numerous' or 'more numerous' are less biologically restricted but in practice are respectively less and more common in average ego networks. All else equal, the structure of kinship and friendship networks would lead us to expect that the most potentially numerous social relations (friends, cousins, nieces/nephews, and aunts/uncles) would be the largest source of LKDs, while less numerous and restricted ties (siblings, children, parents, grandparents, and spouses/partners) would be a smaller share of LKDs.

2.2.2. The transplant-related resource relationship distribution

However, it may be that the distribution of ties with transplant-related biomedical resources determines the LDKT relationship distribution more strongly than the social network tie count relationship distribution alone. Here I define transplant-related resources as alters' biomedical characteristics that promote ability to donate a kidney to ego: being free from medical conditions that preclude LKD, and having a compatible blood type with ego.¹ Relationship groups are likely to vary considerably in their transplant-related resource endowments, in two ways. First, older individuals are much more likely than younger individuals to have disqualifying medical conditions (Daw, 2014), so older-generation kin are likely to be disproportionately medically-excluded from donation. Second, relationship groups also vary in their genetic relationship to ego. However, a close genetic relationship may be a two-edged sword: it could increase LKD due to higher probabilities of blood and tissue type compatibility, but it could also decrease LKD due to shared genetic risk for kidney and related diseases (Cecka, 2007; Satko and Freedman, 2005). The correspondence between transplant-related resources and LKD could differ by race due to substantial race disparities in LKD-disqualifying medical conditions and the prevalence of the *APOL1* genotype in the U.S. non-Hispanic Black population, which statistically increases kidney disease risk

¹ A number of conditions potentially prevent living kidney donation. The exact rules differ across transplant programs, but recent working group recommendations advocate considering several factors, including blood/tissue compatibility; health behaviors and psychosocial characteristics; kidney, cardiovascular, infectious disease, inflammatory, cancer, and metabolic health; family history; and pregnancy (Lentine et al., 2017).

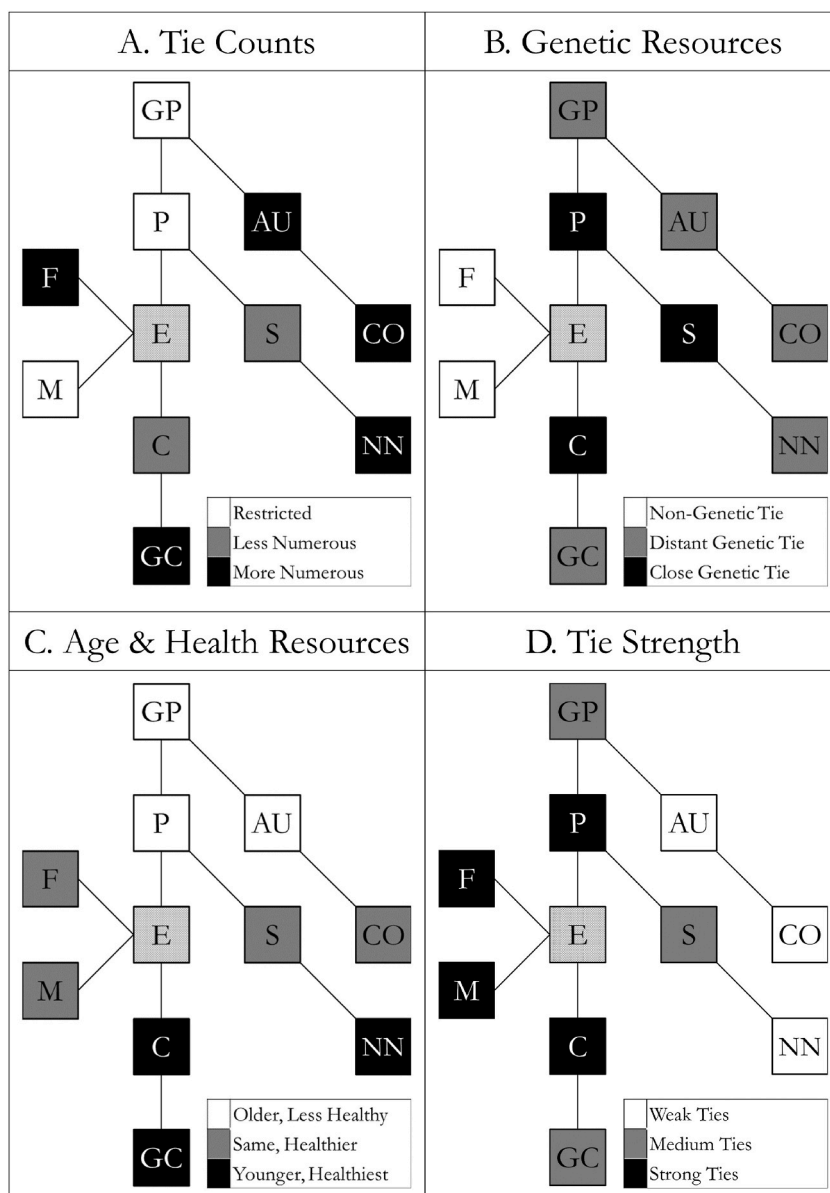


Fig. 1. Conceptual diagram of social capital dimensions' relationship distribution. *Note:* GP = Grandparent, P=Parent, AU = Aunt/Uncle, F=Friend, E = Ego, S=Sibling, CO=Cousin, M = Spouse/partner, C=Child, NN=Niece/Nephew, GC = Grandchild.

and is now commonly part of the LKD evaluation process (Deere and Ferdinand, 2020; Gordon et al., 2019; USRDS. US, 2020). In contrast, although women are less likely than men to develop end-stage kidney disease, they have similar rates of LKD-disqualifying medical conditions, and may often be inhibited from receiving LKDs from their children and their biological fathers due to pregnancy-induced immunological presensitization (Daw, 2014; USRDS. US, 2020; Bromberger et al., 2017).

Panel B in Fig. 1 depicts the structure of expected genetic relationships across relationship groups. Although members of the same relationship group vary in their genetic relationship (since kinship ties can be forged through biological reproduction, marriage/partnership, and/or adoption), I expect that parents, siblings, and children will have the closest average genetic relationship to transplant candidates, while extended kin have more distant genetic relationships, and friends and spouses/partners have no direct genetic relationship. Panel C in Fig. 1 depicts the hypothesized relationship distribution of age and health, which I group together because the hypothesized distribution is the same. Older-generation ties are hypothesized to be substantially less likely to be healthy enough to donate than same-generation ties, who are in turn less likely to be healthy enough to donate than younger-generation ties.

2.2.3. The strong tie relationship distribution

Relationship groups differ substantially in their average tie strength, and tie strength differences likely modify egos' access to

alters' situationally-relevant resources. For instance, previous research on the general population finds significant differences in relationship groups' rates of contact and emotional closeness (Fischer, 2011; Swartz, 2009). For the transplant population, writing long before the current range of donor relationships was permitted (Spital, 1989), Fox and Swazey (1974) anticipated some of the cultural logic that may lie behind the high concentration of living kidney donation among nuclear family members. For instance, they wrote that parents of a young person needing a transplant were the most likely to donate due to the strong parent-child bond prior to children's marriage, and that, although they were not eligible at the time, spouses would be the most culturally-appropriate donors for married recipients (pp. 21–22). However, they also wrote that culturally, children were less likely to donate to parents than the reverse, and that siblings' donation rates were likely to be suppressed by parental or spousal interference, conjectures that are belied by these groups' very high shares of living kidney donors today. Most famously, Fox and Swazey (1992) wrote of "the tyranny of the gift" in which the unreciprocable nature of living kidney donation violates Mauss's (Mauss, 2000) core tenets of gift exchange: to give, receive, and reciprocate. As a result, "the giver, the receiver, and their families may find themselves locked in a creditor-debtor vise that binds them to one another in a mutually fettering way" (p. 40). The implication is that only the strongest social ties are able to reliably withstand the strain of such a symbolically significant, unreciprocable gift. Contemporaneously, Simmons and colleagues (Simmons et al., 1987) extended many of Fox and Swazey's conjectures empirically, and found that close relationships were more likely to discuss donation, volunteer to donate, and be less ambivalent about donation (p. 161, 189). Donors in their studies frequently agreed that donation was primarily a nuclear family obligation (p. 164), particularly a parental one, and that friends would be less likely to donate than family members (p. 165). Today, research finds that patient-alter pairs with recent histories of instrumental support are more likely to take concrete steps toward LKD (Gillespie et al., 2020). There is little reason to think that the strong tie relationship distribution or its LKD consequences differs significantly by race. In contrast, women's aggregate tie strength relationship distribution may differ from men's, as women often serve the role of 'kin keeper' in family networks, have closer average relationships with their children than men, and often have more intensive relationships with their friends (Felmlee and Muraco, 2009; Hagestad, 1986; Shanahan et al., 2007).

Panel D of Fig. 1 depicts the hypothesized tie strength relationship distribution. Spousal/partner ties, parent-child ties, and friends are hypothesized to be adults' strongest average ties, with high access to their resources; followed by siblings, grandparents, and grandchildren with medium tie strength; and aunts/uncles, cousins, and nieces/nephews with low tie strength. All else equal, I expect that the LKD relationship distribution would assume a similar form.

3. Data and methods

3.1. United network for organ sharing standard transplant analysis and research (UNOS STAR) dataset

I characterize the LKD relationship distribution in administrative data using the UNOS STAR files. UNOS is the government contractor that administers the organ transplantation system, and these data describe all legal kidney transplant candidates, recipients, and donors in the U.S. since 1987. I subset these data to completed living donor kidney transplants occurring between 2008 and 17 because this was a period of relatively stable, slow decline in LKD rates before more recent upheavals. These years also encompass the time period in which the survey data was collected (described below).

The *LKD relationship to the transplant candidate* variable is coded as follows in UNOS STAR. There are six "biological, blood related" relations: parent, child, identical twin, full sibling, or half sibling, and other blood related relative (with an open-ended text field to elaborate). There are three non-biological tie categories as well: spouse, life partner, or other unrelated donor (with an open-ended text field to elaborate). The open-ended text fields were used with high clarity and consistency (albeit with varying levels of detail and spelling, leading to 795 distinct values), enabling me to recode these categories into more detailed relationship groups with few exceptions.² Accordingly, I recode LKDs in this time period into one of 11 categories, with biological, step-, by-marriage, and adoptive relations collapsed within each relationship category where relevant. Each relationship is characterized as the donor's relationship to the recipient (e.g., the donor is the parent of the recipient). These recoded relationships include the recipient's *parent, child, sibling, spouse/partner, grandparent, grandchild, aunt/uncle, niece/nephew, cousin, other family* (including in-laws and unspecified family members), and *friends* (defined to include any established non-family relationship such as friends, co-workers, and acquaintances).

There were 51,143 completed living kidney donations to recipients age 20 and older 2008–17.³ Of these, 28,005 are recoded based on the original categories into one of these 11 relationships. Another 15,089 are recoded from 'other' relationship fields into these 11 categories on the basis of open-ended text relationship codes, for a total of 43,094 analyzed LKD relationships (84.3% of the total during this time period). The remaining 8049 LKDs were excluded from the analysis, including 4891 paired exchange transplants (in which two donors donate to each others' intended recipients; UNOS provides no additional relationship details), 1765 anonymous donations, 1165 living/deceased donations (in which one donates to the waiting list to prioritize one's intended recipient for deceased donor kidneys; UNOS provides no additional relationship details), 204 LKDs with an informative open-ended text response that did not fit one of the 11 categories (e.g., strangers), 2 LKDs coded as unknown, 1 domino transplant (where the donor donates their kidney and receives another), and 21 uninformative open-ended text codes.

The LKD relationship distribution is statistically adjusted for candidate race/ethnicity, gender, and age by subsetting the UNOS

² Stata code to reproduce this relationship distribution will be made available to interested readers upon request.

³ 20 was used as the minimum age in the UNOS analysis because IRB restrictions prevented recruiting 18–19 year olds at the ENaCT data collection site.

data to candidates receiving living donor kidney transplants 2008–17 with donor relationships of interest, then predicting donor relationship using a multinomial logit model as a function of candidate race, age, and gender. Average probabilities of each donor relationship are then calculated using the `-margins-` command in Stata 17.0. This procedure only slightly modified the distribution but is presented for consistency with the survey analysis described below.

3.2. Ego networks among candidates for transplant (ENaCT)

Data for the ENaCT study were collected between May and December 2015. ENaCT is a convenience sample of 73 kidney transplant candidates age 20 and older at a single, large transplant center in the United States. Participants were recruited, consented, and surveyed following a nurse-led informational session on kidney transplantation that all early-stage transplant candidates at the center attended before undergoing medical evaluations. Those who wished to participate in the survey were invited to remain in the conference room for full informed consent and to complete the survey, with the aid of a research team member or their companions upon request. Participants were compensated \$10 for their participation. The survey included three major elements: (1) a series of *standard survey* questions about their own characteristics; (2) an *ego network name generator* for each relationship category examined (capturing parents, children, siblings, spouses/partners, grandparents, grandchildren, aunts/uncles, nieces/nephews, cousins, other family, and friends age 20 or older); and (3) a *name interpreter* in which respondents answered a series of 9 questions about each alter (covering demographic characteristics, social and genetic relationships, and donation-related attributes). The survey was self-administered in-person, but a companion or research team member could assist upon request. For this analysis, I removed 29 ex-spouse/partner ties from the analytical sample, dropped 8 alters with inconsistent values across relationship survey items, and restrict the dataset to candidates with complete race information (dropping 1 candidate and 26 candidate-alter pairs). The resulting analytical dataset consists of 72 transplant candidates and 1548 network ties.

Relationship to the candidate is coded based on the name generator section where the alter was listed. Just as in UNOS STAR, alter-ego relationships are characterized from the transplant candidate's perspective, characterizing the relationship of each alter to ego (e.g., alter is ego's parent). However, as a single-center sample of kidney transplant candidates, this sample is far less representative than the administrative data to which I compare it, and I therefore statistically adjust its estimated relationship distributions for candidate race/ethnicity, gender, and age. To do so, I transformed the ENaCT data into a candidate-relationship group level dataset, with the percentage of alters with each characteristic in the relationship group expressed as the dependent variable. These percentages were then regressed on relationship group and candidate race, age, and gender, and the average predicted percentages by relationship group were calculated using the `-margins-` command in Stata 17.0.

To better understand the LKD relationship distribution, I compare each relationship group's share of all network ties, share of ties with transplant-related biomedical resources, and share of strong ties with the same relationship group's share of LKDs in the UNOS STAR data. Each ENaCT measure is calculated as

$$p_r = 100 \times \frac{\sum c_r}{\sum c}$$

where r indexes relationships, c is a dichotomous alter characteristic, and p_r is relationship r 's share of ties with characteristic c . For a relationship group's *share of all network ties*, the characteristic is simply their existence in the analytical sample. For a relationship group's *share of ties with transplant-related resources*, three characteristics were coded for each alter, indicating younger age ($=1$ if ≤ 50 years old, $=0$ otherwise),⁴ perceived health ($=1$ if candidate believes they are healthy enough to donate a kidney, $=0$ if not or don't know),⁵ and predicted probability of blood type compatibility. Blood type compatibility probability is calculated using a combination of ENaCT and UNOS STAR data. In this calculation, the implied genetic relationship between ego and alter in ENaCT,⁶ the race-specific ABO allele distribution in UNOS, and the rules of blood type compatibility are used to calculate the probability that each alter has a compatible blood type with the transplant candidate using well-established population genetic formulas (Kanter and Hodge, 1990), and each alter-ego relationship is simulated 10,000 times to convert these values to dichotomous characteristics ($=1$ if simulated compatible, $=0$ otherwise; see appendix for full calculation details). Each relationship group's share of compatible ties is then calculated on this simulated dataset. Finally, for each relationship group's *share of strong ties*, two dichotomous variables were coded: high communication frequency ($=1$ if communicate at least weekly, $=0$ otherwise),⁷ and high relationship closeness ($=1$ if extremely close, $=0$ otherwise).⁸

⁴ For each alter, the candidate is asked, "What is this person's age (best guess)?" The response options include 20–30, 31–50, 51–70, and 71+.

⁵ For each alter, the candidate is asked, "To your knowledge, do you think they are healthy enough to donate a kidney?" Response options include "No," "Yes," and "Don't Know." Notably, this survey was given immediately after the candidates were given a presentation on living donor kidney transplants, including donor eligibility rules, so it is likely they were reasonably knowledgeable about what conditions would or would not be disqualifying.

⁶ For each relationship group other than spouses and friends in ENaCT, respondents are asked "How are you related to this person?" with the response options "Blood relative" and "Other."

⁷ For each network member, the transplant candidate is asked, "How often do you talk to this person?" Response options included "Once a year or less," "2–11 times a year," "Once a month," or "Once a week or more."

⁸ For each network member, the transplant candidate is asked, "How close are you to this person?" The response options included "Not Very," "Somewhat," "Very," and "Extremely."

Transplant candidate *race* and *gender* are measured by candidate self-report in the standard survey component of the study. Because only three Latino patients and no Asians, Pacific Islanders, Native Americans, or Alaska Natives participated in this survey, I restrict the race- and gender-stratified analyses to non-Hispanic White and Black patients only. Gender is characterized as either male or female by candidates' self-identification.

3.3. Statistical analysis

In the full sample analysis, I compare each relationship group's percentage of all LKDs in UNOS STAR (dependent variable) to each relationship group's percentage of all ties, percentage of ties with each transplant-related resource, and percentage of ties with each strong tie measure in ENaCT (independent variables). The relationship between the dependent variable and each independent variable is presented as a series of scatterplots. In these scatterplots, each point is a relationship group. Since both variables are on the same (percentage) scale, each scatterplot point's proximity to the 45-degree line on the scatterplot may be interpreted as follows: (a) points exactly on the 45-degree line indicate that their share of LKDs are proportional to their share of the independent variable in candidates' social networks (*proportional utilization*); (b) points in the upper-left triangle indicate that the relationship group's share of LKDs exceeds their share of the independent variable in candidates' social networks (*relative overutilization*); and (c) points in the lower-right triangle indicate that the relationship group's share of LKDs falls short of their share of the independent variable in candidates' social networks (*relative underutilization*).

A simple example may further clarify: Suppose that ENaCT measures 1000 network ties among 100 candidates, and there are only four relationship groups — A, B, C, and D —with 250 ties (25%) apiece. Additionally, suppose that 40% of LKDs in UNOS were from group A, 25% each were from B and C, and 10% were from D. In this scenario, compared to their share of all ties, group A was *relatively overutilized* (40% of LKDs in UNOS exceeds 25% of all ties in ENaCT); groups B and C were *proportionally utilized* (25% of LKDs equals 25% of all ties); and group D was *relatively underutilized* (10% of LKDs is less than 25% of all ties). To illustrate the comparison to transplant-related resources, suppose that the following number of ties were rated healthy enough to donate in ENaCT: 120 in group A; 100 in group B; 50 in group C; and 30 in group D, for a total of 300 healthy ties. Group A is 40% (= 120/300) of ties healthy enough to donate, group B is 33% (= 100/300), group C is 17% (= 50/300), and group D is 10% (= 30/300). Thus, relative to their share of healthy ties, group A is proportionally utilized (40% of LKDs in UNOS equals 40% of healthy ties in ENaCT), group B is relatively underutilized (25% of LKDs is less than 33% of healthy ties), group C is relatively overutilized (25% of LKDs exceeds 17% of healthy ties), and group D is proportionally utilized (10% of LKDs equals 10% of healthy ties). This same interpretative approach applies to each relationship group's share of tie counts, each transplant-related resource, and each strong tie indicator.

To determine which independent variable best accords with the LKD relationship distribution, I use the mean absolute deviation (MAD) as a measure of model fit, where deviations are defined relative to the 45-degree ($y = x$) line in the scatterplot. This statistic is calculated as

$$MAD_c = \frac{\sum_r |y_{rc} - \hat{y}_{rc}|}{R}$$

where MAD_c is the mean absolute deviation when the characteristic c is used as the independent variable, r indexes relationship groups, R is the total number of relationship groups (=11), y_{rc} is the observed percentage of LKDs in relationship group r with characteristic c , and $\hat{y}_{rc}$ is the relationship group r 's share of ties with characteristic c (and thus the predicted share of LKDs on the 45-degree line). Smaller MAD values indicate better fit (i.e., smaller average residuals). I calculate MAD rather than the more traditional r^2 because residuals are calculated compared to a fixed 45-degree line rather than an estimated one (as is typical with r^2), and MAD has the virtue of easy interpretation (i.e., if $MAD = 5$, the average scatterplot point is 5 vertical units from the 45-degree line).

In order to assess each model's proportional reduction in error compared to a null model, I also calculate MAD_{null} using the same equation but substituting the grand mean percentage of LKDs in each relationship group ($\bar{y} = (100/11)$) for $\hat{y}_r$. The purpose is to assess each model's fit not only relative to other models but also relative to random guessing; if a model cannot improve fit over random guessing, it is a poor model. The proportional reduction in error of each model compared to this null model is then calculated as

$$PRE_c = 100 \left(\frac{MAD_{null} - MAD_c}{MAD_{null}} \right)$$

PRE_c scales from 0 to 100, with higher PRE_c values indicating better explanatory value.

Finally, I conduct three robustness checks. First, in order to determine whether the study's conclusions hold within key population subgroups, each model is re-estimated using race- or gender-stratified data in both UNOS STAR and ENaCT. Race- and gender-specific LKD relationship distributions are also calculated to provide important context for these assessments. Second, each model's MAD and PRE are recalculated with each relationship group omitted one at a time to determine whether a single relationship group determines the model fit. Third, because I am comparing single-center data to national data, I assess the difference in the relationship distribution of the focal state relative to other states in UNOS STAR to contextualize these findings.

4. Results

4.1. Sample descriptions

4.1.1. UNOS STAR living kidney donor relationship distribution

Table 1 shows the distribution of LKD relationships in UNOS STAR, adjusted for candidate race/ethnicity, age, and gender. There is enormous relationship group variation in the share of LKDs. Nuclear family ties account for the majority of LKDs: siblings are 25% of LKDs, children are 18%, spouses/partners are 16%, and parents are 6%, for a total of 65%. Friends account for an additional 20%, leaving only 15% of examined LKDs for extended kin. Among non-nuclear family, grandparents (0.03%) donate nearly no kidneys to their adult grandchildren, while cousins (4%), nieces/nephews (3%), aunts/uncles (2%), and grandchildren (0.3%) are donors in a small percentage of cases. Other family members such as in-laws and uncategorized family members account for the remaining 7% of analyzed ties.

4.1.2. ENaCT network sample description

Table 1 also describes the structure and characteristics of ENaCT's network data, adjusted for candidate race, age, and gender. As expected, cousins (N = 344, mean = 4.8, 20% of all ties), siblings (N = 236, mean = 3.3, 17%), aunts/uncles (N = 234, mean = 3.3, 14%), friends (N = 216, mean = 3.0, 13%), and nieces/nephews (N = 195, mean = 2.7, 12%) were the most numerous reported ties, with comparatively few spouses/partners (N = 38, mean = 0.5, 3%), grandchildren (N = 20, mean = 0.3, 1.5%), or grandparents (N = 18, mean = 0.3, 2.2%) reported. The right-hand side of Table 1 depicts each relationship group's share of ties with each transplant-related biomedical resource or tie strength indicator in candidate networks, adjusted for candidate race, age, and gender. The correspondence of these values to each relationship group's share of LKDs in UNOS STAR data will be explored in the scatterplot analyses below and are presented here for reference. Appendix Table A describes the candidate race and gender distribution in ENaCT. 33% were non-Hispanic White, 60% were non-Hispanic Black, 5% were Latino/a, 40% of the respondents were men, and 59% were women.

4.2. Aggregate scatterplot analysis

Fig. 2 provides the results of the aggregate scatterplot analyses. In each subgraph, the y-axis represents the adjusted percentage of LKDs in each relationship group in UNOS STAR, plotted against the percentage of ENaCT alters in the same relationship groups with the independent variables. Table 2 lists the residuals for each model in Fig. 2. As shown in the bottom rows of Table 2, the null model predicting that each relationship group will have an equal share of LKDs ($100/11 = 9.9\%$) has a MAD of 7.7, so an important question for each scatterplot in Fig. 2 is whether its fit to the data improves upon this fit. Panel (A) in Fig. 2 shows that network structure alone is a weak fit for the relationship distribution of LKDs. The MAD is 7.6, which is just 1.4% lower than the null model MAD of 7.7, suggesting little explanatory value for this variable. However, the relationship group residuals relative to this baseline are nonetheless informative. Compared to the network tie distribution, the most relatively overutilized LKD relationships are spouses/partners (16.1% of LKDs, 3.2% of network, residual = 12.9), children (18.3% of LKDs, 7.3% of network, residual = 11.0), siblings (24.6% of LKDs, 17.0% of network, residual = 7.6), and friends (19.7% of LKDs, 12.5% of network, residual = 7.2). Conversely, cousins (3.6% of LKDs, 20.1% of network, residual = -16.5), aunts/uncles (1.6% of LKDs, 14.3% of network, residual = -12.7), and nieces/nephews (2.8% of LKDs, 12.1% of network, residual = -9.3) are the most relatively underutilized LKD relationships compared to their share of all ties. Other relationships are approximately proportionally utilized as LKDs.

The remaining panels in Fig. 2 perform the same analysis for members with each biomedical resource (age ≤ 50, healthy, simulated to have a compatible blood type) and two measures of tie strength (talk weekly and closeness), and Table 2 also depicts the relationship

Table 1

Percentage distribution of living kidney donor relationships to recipient and composition of transplant candidate networks.

Relationship to Candidate	Abb	UNOS % LKDs	ENaCT Network Structure			ENaCT Percentage of __ Ties				
			N(total)	N(mean)	% of All Ties	Biomedical Resources			Tie strength	
						Age ≤ 50	Healthy	Compatible	Talk Weekly	Ext Close
Parents	P	6.1	63	0.9	5.3	1.6	2.3	5.9	10.8	15.0
Children	C	18.3	99	1.4	7.3	14.1	12.3	8.0	12.8	18.5
Siblings	S	24.6	236	3.3	17.0	15.5	15.9	19.3	22.3	20.4
Spouses/Partners	M	16.1	38	0.5	3.2	3.0	5.1	2.8	6.8	11.2
Grandparents	GP	0.0	18	0.3	2.2	0.0	0.0	2.1	1.3	1.8
Grandchildren	GC	0.3	20	0.3	1.5	2.6	2.6	1.5	2.0	2.4
Aunts/Uncles	AU	1.6	234	3.3	14.3	1.5	4.7	14.3	6.4	4.5
Nieces/Nephews	NN	2.8	195	2.7	12.1	21.6	19.1	12.0	6.2	7.3
Cousins	CO	3.6	344	4.8	20.1	22.7	19.5	19.0	6.1	5.5
Other Family	OF	6.8	85	1.2	4.6	4.7	5.9	4.1	4.1	2.1
Friends	F	19.7	216	3.0	12.5	12.6	12.7	11.1	21.1	11.1

NOTE: All percentage figures describe the percentage of ties with a given trait in each relationship category (i.e., not the percentage of ties in each relationship category with that trait). All percentage figures are statistically adjusted for transplant candidate age, race, and gender; thus, the “% of All Ties” column, which is adjusted, does not exactly match the percentages implied by the N(total) column, which is not adjusted.

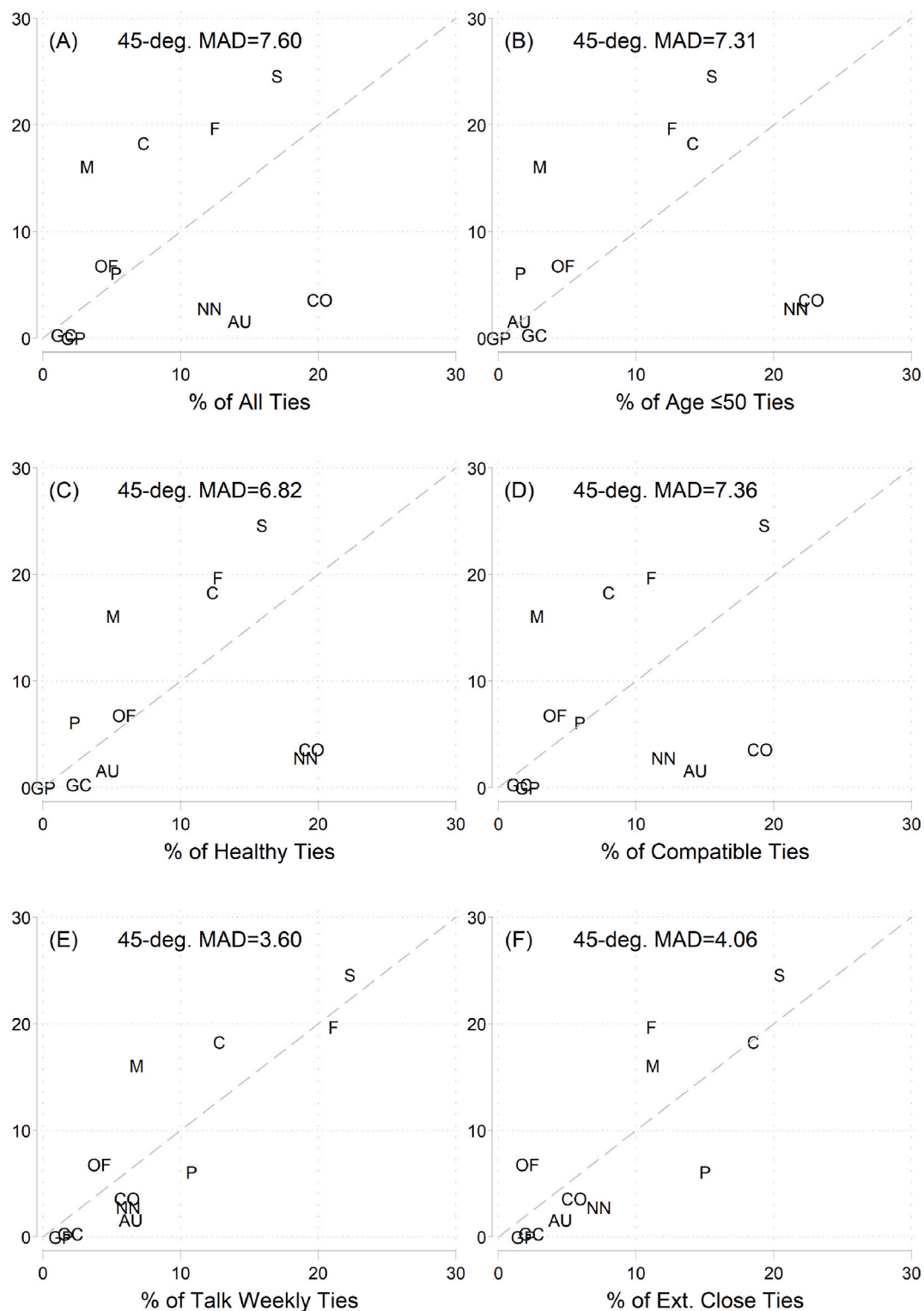


Fig. 2. Percentage of Living Kidney Donors versus Percentage of Candidate Social Networks, by Relationship and Trait. NOTE: The y-axis is % of living donors by relationship type in UNOS STAR data; the x-axis is % of alters with indicated trait in ENaCT data. 45-deg. MAD is the mean absolute value of the residual from the 45-degree line. Abbreviations: P=Parent, C=Child, S=Sibling, GP = Grandparent, GC = Grandchild, AU = Aunt/Uncle, NN=Niece/Nephew, CO=Cousin, OF=Other Family, F=Friend.

Table 2
Scatterplot residuals.

Relationship	% LKDs	Null Model	Network Structure	Biomedical Resources			Tie strength	
		Mean %	% All Ties	% Age ≤50 Ties	% Healthy Ties	% Compatible Ties	% Talk Weekly Ties	% Ext. Close Ties
Parents	6.1	−3.0	0.8	4.5	3.8	0.2	−4.7	−8.9
Children	18.3	9.2	11.0	4.2	6.0	10.3	5.5	−0.2
Siblings	24.6	15.5	7.6	9.1	8.7	5.3	2.3	4.2
Spouses/Partners	16.1	7.0	12.9	13.1	11.0	13.3	9.3	4.9
Grandparents	0.0	−9.1	−2.2	0.0	0.0	−2.1	−1.3	−1.8
Grandchildren	0.3	−8.8	−1.2	−2.3	−2.3	−1.2	−1.7	−2.1
Aunts/Uncles	1.6	−7.5	−12.7	0.1	−3.1	−12.7	−4.8	−2.9
Nieces/Nephews	2.8	−6.3	−9.3	−18.8	−16.3	−9.2	−3.4	−4.5
Cousins	3.6	−5.5	−16.5	−19.1	−15.9	−15.4	−2.5	−1.9
Other Family	6.8	−2.3	2.2	2.1	0.9	2.7	2.7	4.7
Friends	19.7	10.6	7.2	7.1	7.0	8.6	−1.4	8.6
Mean Absolute Deviation	–	7.7	7.6	7.3	6.8	7.4	3.6	4.1
Proportional Reduction in Error	–	–	1.4	5.2	11.6	4.5	53.3	47.3

NOTE: % LKDs is the percentage distribution of living kidney donor relationship types. Values in the main box are the relationship residuals compared to the 45-degree line expressed. The second-to-bottom row is the mean absolute deviation for each scatterplot. The bottom row is the percentage of the null model MAD eliminated in the model in question. In each row, the lowest residual or MAD is highlighted in bold (ignoring the mean % column, which is presented for comparison only). Cells shaded blue are positive residuals; cells shaded red are negative residuals. The lowest MAD and relationship-specific absolute deviation is highlighted in bold.

residuals associated with each comparison. Panels (B)–(F) in Fig. 2 share some key commonalities. In all of these comparisons, spouses/partners and siblings are relatively overutilized as LKDs, while nieces/nephews and cousins are relatively underutilized. For less consistent patterns, children are relatively overutilized in all models except the extremely close ties model, where they are slightly relatively underutilized. Parents are relatively overutilized relative to their share of biomedical resources except for compatible ties (where they are proportionally utilized), and proportionally underutilized relative to both tie strength indicators. Aunts/uncles are relatively underutilized in most models, but not in the age≤50 relationship distribution.

Together, Fig. 2 and Table 2 shows that the relationship distribution of talk weekly ties (panel E, MAD = 3.6, PRE = 53.3%) best fits the LKD relationship distribution, followed by the distribution of extremely close ties (panel F, MAD = 4.1, PRE = 47.3%). By comparison, the percentage of compatible ties (panel D, MAD = 7.4, PRE = 4.5%), ties age≤50 (panel B, MAD = 7.3, PRE = 5.2%), and healthy ties (panel C, MAD = 6.8, PRE = 11.6%) do not perform meaningfully better than the null model. Thus, in aggregate, tie strength appears to have the potential to better explain the LKD relationship distribution than transplant-promoting biomedical resources or tie counts.

However, there is considerable relationship group variability in how well each factor accords with their share of LKDs. At the relationship group rather than aggregate level, the talk weekly model yields the smallest absolute deviation for siblings, nieces/nephews, and friends only. In contrast, the smallest absolute deviation for parents and grandchildren is their percentage of compatible ties; for children, spouses/partners, and cousins it is their percentage of extremely close ties; for grandparents and aunts/uncles it is percentage of ties age≤50 (tied with healthy ties in the case of grandparents); for other family members it is their percentage of healthy ties. In total, tie strength variables minimize the residual for 6 relationship groups and transplant-promoting biomedical resources minimize the residual for 5 relationship groups.

To summarize, these findings suggest that: (1) Parents' and grandparents' moderate and low shares of LKDs are attributable to their strong ties but limited by their low health- and age-related biomedical resources; (2) Children's, spouses/partners', and siblings' high share of LKDs are best attributed to their strong ties (along with high age and health transplant-promoting biomedical resources for children); (3) Grandchildren's low share of LKDs is best attributed to their low share of all and ABO-compatible ties; (4) Aunts/uncles' low share of LKDs are best attributable to a combination of their low biomedical resources and tie strength; (5) Nieces/nephews' and cousins' low shares of LKDs are best attributed to their weak ties (despite their high shares of biomedical resources); (6) Other family members' moderate share is best attributed to their good health; and (7) Friends' high share of LKDs are best attributed to their high share of talk weekly ties.

4.3. Robustness checks

4.3.1. Race- and gender-stratified analyses

The online appendix presents the results of the race- and gender-stratified analyses (Tables B–C, Figures A–D). The LKD relationship distributions did differ appreciably by race (Table B), as White transplant candidates had higher LKD shares from friends, other family, and parents, while Black transplant candidates had higher shares from children. The LKD relationship distribution is very similar for men and women recipients except for spouses/partners. The results of the scatterplot analyses show that the primary results hold for each race and gender group (Figures A–D, Table C). The only small difference was that for the Black patient comparison, the emotional

closeness model fit slightly better than did the talk weekly model ($MAD = 4.43$ vs. 4.87 , Table C).

4.3.2. Sensitivity to individual relationship groups

Online appendix Table D reports the MAD statistics for Fig. 2 (on the 'None' row) and a series of supplemental tests in which I dropped one relationship from the scatterplot and then once again calculated the residuals and MAD. The purpose of this is to check whether a single relationship group was driving the conclusions in the primary analyses. Table D shows that this is not the case — the percentage of talk weekly ties had a lower MAD than any other test, with the exception of friends. When friends were dropped, the extremely close ties model had the lowest MAD, but as this is another close ties indicator, this does not affect the theoretical conclusions of the analysis.

4.3.3. Living donor relationship distribution by state

Online Appendix Table E presents the results of a cross-tabulation of the living kidney donor relationship distribution comparing the focal state (where ENaCT data were collected) to other states 2014–16. The results of a chi-squared test are statistically significant ($p = 0.01$), but the substantive differences are modest. Only the number of child (9.6% in focal state versus 17.2% in other states) and friend (29.4% versus 20.8%) donors in the focal state differed appreciably from the frequency expected from the national data. The national data are used in this analysis due to small sample size concerns in the focal state alone, but these moderate differences provide important context.

5. Conclusions

In the network perspective, social capital is the sum of latent resources inhering in ego's social connections, where ego's ability to mobilize these resources into social support is modified by ego-alter tie strength. In this perspective, the relationship distribution of mobilized social support may be determined by the relationship distribution of network tie counts, relevant resources, and tie strength. Applying this perspective, I assessed how the relationship distribution of each social capital dimension accorded with the LDKT relationship distribution. I found that the tie strength relationship distribution accorded most tightly with the LKD relationship distribution. This conclusion was upheld in race- and gender-stratified analyses, and is robust to the omission of each relationship group.

It is likely intuitive to most social scientists that the closest ties would be the most common LKDs, as this act is often construed as a major personal sacrifice made on behalf of a loved one which cannot be reciprocated. What is perhaps more surprising is that the LKD relationship distribution corresponds so poorly to the transplant-related biomedical resource relationship distribution. Unlike many other forms of social support, LKD is subject to robust medical institutional gatekeeping, as transplant centers enforce clear guidelines on who is and is not permitted to become an LKD. Many willing potential donors are thereby prevented from donating, which might have been expected to reduce the salience of the tie strength relationship distribution and increase the salience of the transplant-related biomedical resource distribution. However, I emphasize that this result is an aggregate finding, not a pair-level one; which specific individuals within a relationship group actually donate a kidney will still be heavily influenced by their health and compatibility with the recipient. Moreover, the conclusion that the tie strength relationship distribution more strongly corresponds with the LKD relationship distribution than the transplant-related biomedical resource or social network tie count relationship distributions is an aggregation across relationship groups. In several cases, a specific relationship group's share of LKDs best accords with a different social capital dimension. For instance, parents' share of LKDs more strongly accords with network structure and biomedical resources than it does tie strength, and aunts/uncles' low share of LKDs is proportional to their low share of ties under age 50. It is simply the average error that is minimized as a function of tie strength.

These results should be interpreted with some caution, and further work is needed to confirm these findings. Although the number of network ties examined in ENaCT is substantial, the number of transplant candidates in this survey is moderate and limited to a convenience sample of patients at a single transplant center. Moreover, the mode of analysis in this paper is descriptive and indirect, looking for clues to the observed LKD relationship distribution nationwide in a survey unconnected to these outcomes. However, I argue that these comparisons are informative as the basis for future research in which these outcomes are more directly linked, for which data does not currently exist. Furthermore, data limitations in UNOS STAR prevent identification of living donors' relationships to their intended recipient in paired exchange and living/deceased donation arrangements, and these donors are likely to be disproportionately non-genetically-related relationship groups (Flechner et al., 2018) such as spouses/partners, friends, in-laws, adoptees, and relatives by marriage, which may bias these estimates to an uncertain degree.

Even with these cautions in mind, however, I argue that these findings are important for understanding and intervening to promote LDKT. In terms of these important practical concerns, I believe the relationship-specific rather than average residuals are most informative. In particular, nieces/nephews' and cousins' large negative residuals in Table 2 suggest that these regions of transplant candidates' networks are vastly underutilized as LKDs from a medical perspective, and that interventions to promote LKD among these groups offer an important avenue to those seeking to promote higher LDKT rates. Although ethically those who are well-informed about LKD's risks and benefits who do not wish to donate should not be further pressed to do so, it may be that many such individuals would respond differently to different presentations of the issue. For instance, the dominant framing of organ donation in the U.S. is as "the gift of life," which, while apt, conjures the image a deeply intimate act, akin to childrearing or intensive caregiving. Typically, only the closest social relations provide these forms of social support, and it may be that this framing is effective for strong ties but less so for weaker ties. Consider the social construction of other prosocial behaviors that involve making one's organs and tissues available for others' benefit. Blood, egg, and bone marrow donation are all acts that members of the general population routinely engage in to benefit total strangers, and are framed as altruistic but non-intimate acts (Titmuss, 2018; Almeling, 2011; Healy,

2006; Simmons et al., 1993). For example, bone marrow donation promotion campaigns exhort the public not to give the gift of life, but to “be the match” that others need.^{9,10} As another alternative, sperm and plasma donation are commonly framed as ways to help others while earning easy cash (Almeling, 2011; Healy, 2006). Even birth surrogacy, a form of social support just as intimate as kidney donation (and with similarly low mortality risks), is often a paid arrangement between strangers (Jacobson, 2016). Building on Fox and Swazey’s (Fox and Swazey, 1974) concept of the tyranny of the gift, it may be that the ‘gift’ framing that dominates organ donation promotion efforts is particularly poorly suited for promoting living kidney donation from less intimate relations, since unreciprocable gift exchange typically requires a very strong social tie to be sustained. LKD promotion campaigns mimicking the bone marrow or plasma donation framings, for instance by promoting empathy for transplant candidates or even framing donation as a way to do good while effectively receiving paid leave from work (due to recently-expanded federal policies designed to make LKD a completely financially-neutral act),^x may prove more effective at promoting LKD from these underutilized relationship groups.

Additionally, I believe the research strategy adopted in this paper could have utility in other applications as well. The extension to others forms of social support research is straightforward when data is collected on social network ties engaging in a behavior but not on ties who do not, as in the case of important matters discussion name generators (Marsden, 1987). However, the exact approach taken here would not be the most appropriate one for studying the determinants of important matters discussions, since one could simply ask whether ego discusses important matters with each alter when collecting the more comprehensive social network data. Instead, this approach would be most appropriate when data on relationships of someone engaging in some behavior is available but only in isolation from network data in a manner not easily replicated through survey measures. Perhaps data on who unmarried individuals list as their emergency contact at work or in medical care or who are named beneficiaries on different types of financial accounts would be appropriate applications of this method.

I add that the data limitation addressed here is more general than social support research. As social scientists we frequently encounter datasets where we have data on those who do or are something but no data on those who do not do or are not that thing. In case-control research designs analyzing hospital record or public health data on patients with some affliction of interest, this is handled by treating members of a more general population as controls to which cases are compared to test for differences in exposure rates (Breslow, 1996). Similarly, in certain domestic administrative datasets we have data on foreign-born residents (such as their year of birth, year of immigration, and country of birth), but no information about their co-nationals who did not immigrate. If one wished to study the determinants of immigration to the focal receiving nation, one might compare the characteristics of those remaining in the country of birth to those of immigrants from that country (Dustmann et al., 2012). In both cases, the researcher compares data on those who meet some criterion (presentation with disease, international migration) to the same characteristics measured separately among those in a population at risk of meeting that criterion (in both of these cases, linked geographically). My approach is simply another tool in this toolkit, applied to a much more specific population at risk of an event (alters in transplant candidates’ social networks at risk of living kidney donation), probably with more limited utility. In sum, it is critical to identify not only the relationship groups that typically provide support but also those that do not in order to identify the social and resource barriers to social support mobilization. This is especially true in LKD research, where a surprisingly small share of kidney transplant candidates obtain an LDKT. By seeking the network regions where situationally-relevant resources are abundant but ties are weak, we may be able to meaningfully intervene to promote this life-saving outcome.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssresearch.2023.102888>.

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